

JUnit/Mockito Exercise 3: Mockito Basic Mocking Stubbing

Mockit- Hands-On Exercises

Exercise 1: Mocking and Stubbing

Scenario:

You need to test a service that depends on an external API. Use Mockito to mock the external API and stub its methods.

Steps:

1. Create a mock object for the external API.
2. Stub the methods to return predefined values.
3. Write a test case that uses the mock object.

Solution Code:

```
import static org.mockito.Mockito.*;
import org.junit.jupiter.api.Test;
import org.mockito.Mockito;

public class MyServiceTest {
    @Test
    public void testExternalApi() {
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
        when(mockApi.getData()).thenReturn("Mock Data");
        MyService service = new MyService(mockApi);
        String result = service.fetchData();
        assertEquals("Mock Data", result);
    }
}
```

Exercise 2: Verifying Interactions

Scenario:

You need to ensure that a method is called with specific arguments.

Steps:

1. Create a mock object.
2. Call the method with specific arguments.
3. Verify the interaction.

JUnit/Mockito Exercise 3: Mockito Basic Mocking Stubbing

Solution Code:

```
import static org.mockito.Mockito.*;
import org.junit.jupiter.api.Test;
import org.mockito.Mockito;

public class MyServiceTest {
    @Test
    public void testVerifyInteraction() {
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
        MyService service = new MyService(mockApi);
        service.fetchData();
        verify(mockApi).getData();
    }
}
```

Exercise 3: Argument Matching

Scenario:

You need to verify that a method is called with specific arguments.

Steps:

1. Create a mock object.
2. Call the method with specific arguments.
3. Use argument matchers to verify the interaction.

Exercise 4: Handling Void Methods

Scenario:

You need to test a void method that performs some action.

Steps:

1. Create a mock object.
2. Stub the void method.
3. Verify the interaction.

Exercise 5: Mocking and Stubbing with Multiple Returns

Scenario:

You need to test a service that depends on an external API with multiple return values.

JUnit/Mockito Exercise 3: Mockito Basic Mocking Stubbing

Steps:

1. Create a mock object for the external API.
2. Stub the methods to return different values on consecutive calls.
3. Write a test case that uses the mock object.

Exercise 6: Verifying Interaction Order

Scenario:

You need to ensure that methods are called in a specific order.

Steps:

1. Create a mock object.
2. Call the methods in a specific order.
3. Verify the interaction order.

Exercise 7: Handling Void Methods with Exceptions

Scenario:

You need to test a void method that throws an exception.

Steps:

1. Create a mock object.
2. Stub the void method to throw an exception.
3. Verify the interaction.